

MTH209: Assignment 13

Confidence Interval

Suppose $X_1, X_2, \dots, X_{n_1}$ are iid observations from $N(\mu_1, 5)$ and $Y_1, Y_2, \dots, Y_{n_2}$ are iid observations from $N(\mu_2, 10)$ (note that both variances are known). We are interested in estimating $\mu_2 - \mu_1$ and my choice of estimator is

$$T_n = \frac{1}{n_2} \sum_{i=1}^{n_2} Y_i - \frac{1}{n_1} \sum_{i=1}^{n_1} X_i.$$

The function below creates an exact confidence interval for T_n given data **x** and **y**. Complete the function and submit only the function **assignment13.R**.

```
# x: a vector of the X observations
# y: a vector of the Y observations
# alpha: confidence level of the Confidence Interval
make_CI <- function(x, y, alpha)
{
  ...
  ...
  Lower <- .... # calculate lower end of CI
  Upper <- .... # calculate upper end of CI

  return(c(Lower, Upper))
}
```

NOTE: Only upload the function **make_CI** in **assignment12.R** and nothing else.